SUMMER RESEARCH PROJECT PROPOSAL

Beyond Black-Scholes: Fitting Lévy Processes to Stock Returns

A Frequentist and Bayesian Comparison of Heavy-Tailed Distributional Models

Applied Statistics | S&P 500 Daily Returns | Python

1 Lay Summary

Standard financial models assume that daily stock price movements follow a normal (bell-curve) distribution. In reality, extreme market events—crashes and sharp rallies—occur far more often than this assumption predicts. This project fits more realistic mathematical models, known as **Lévy processes**, to S&P 500 daily returns and tests whether they better capture the risk of rare but severe losses. Using both classical (frequentist) and Bayesian statistical methods, the goal is to quantify precisely how much the standard Gaussian model *underestimates* financial tail risk.

2 Detailed Project Description

2.1 Motivation and Background

The Black-Scholes (1973) framework assumes that the log-return of a stock price S_t over an infinitesimal interval dt evolves as:

$$d(\log S_t) \;=\; \mu\, dt \;+\; \sigma\, dW_t, \tag{1}$$

where W_t is a standard Brownian motion. This implies that daily log-returns $r_t = \log(S_t/S_{t-1})$ are Gaussian: $r_t \sim \mathcal{N}(\mu\Delta t,\, \sigma^2\Delta t)$. However, decades of empirical evidence—spanning the 1987 crash, the 2008 global financial crisis, and the March 2020 COVID shock—show that real equity returns exhibit:

- **Excess kurtosis**: extreme moves occur far more often than a Gaussian predicts.
- **Negative skewness**: large losses are more probable than equivalent gains.
- **Semi-heavy tails**: the tail decay is slower than exponential but faster than power-law.

Lévy processes offer a natural and mathematically principled generalisation. A Lévy process X_t satisfies: (i) independent increments, (ii) stationary increments, and (iii) càdlàg sample paths. By the *Lévy–Khintchine theorem*, any such process is characterised by its triplet $(b,\, \sigma^2,\, \nu)$ — drift, Gaussian component, and a Lévy measure ν controlling jump behaviour:

$$\log \mathbb{E}\left[e^{iuX_t}\right] \;=\; t\left(ibu \;-\; \tfrac{1}{2}\sigma^2 u^2 \;+\; \int_{\mathbb{R}\setminus\{0\}} \left(e^{iux} - 1 - iux\,\mathbf{1}_{|x|<1} \right) \nu(dx) \right). \tag{2}$$

2.2 The Model Hierarchy

Four distributional models are fitted and compared to daily S&P 500 log-returns:

Model	Parameters	Key Feature	Lévy Process?
Gaussian (BM)	μ, σ^2	Baseline; thin tails	Yes (trivially)
Student-t	μ, σ, ν	Heavier tails via degrees of freedom ν	No – motivational
Variance-Gamma (VG)	μ, σ, ν, θ	Skewness + excess kurtosis; finite moments	**Yes – primary**
Normal Inverse Gaussian (NIG)	$\alpha, \beta, \mu, \delta$	Flexible semi-heavy tails	Yes – extension

2.3 Frequentist Estimation: Maximum Likelihood

For each model $\mathcal{M}$ with parameter vector $\boldsymbol{\theta}$, the log-likelihood over n i.i.d. observations $\{r_1, \ldots, r_n\}$ is maximised numerically:

$$\hat{\boldsymbol{\theta}}_{\text{MLE}} \;=\; \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log f(r_i;\, \boldsymbol{\theta},\, \mathcal{M}). \tag{3}$$

For the Variance-Gamma model, the PDF involves a modified Bessel function of the second kind K_ν:

$$f_{\text{VG}}(x;\, \mu, \sigma, \nu, \theta) \;=\; \frac{2\, e^{\theta(x-\mu)/\sigma^2}}{\sigma\sqrt{2\pi}\,\Gamma(1/\nu)\,(nu)^{1/\nu}} \left(\frac{|x-\mu|}{\sqrt{2\sigma^2/\nu + \theta^2}} \right)^{1/\nu - 1/2} K_{1/\nu-1/2}\!\left(\frac{\sqrt{2\sigma^2/\nu + \theta^2}\,|x-\mu|}{\sigma^2} \right). \tag{4}$$

Both VG and NIG PDFs are evaluated using `scipy.special` and optimised via `scipy.optimize.minimize` with the L-BFGS-B method. Standard errors are recovered from the inverse Hessian of the log-likelihood at the optimum, yielding asymptotic confidence intervals via the Fisher information matrix:

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) \;\approx\; \left[-\frac{\partial^2 \ell}{\partial \boldsymbol{\theta}\,\partial \boldsymbol{\theta}^\top}\bigg|_{\hat{\boldsymbol{\theta}}} \right]^{-1}, \tag{5}$$

where $\ell = \sum_i \log f(r_i; \boldsymbol{\theta})$ is the log-likelihood.

2.4 Bayesian Extension: Posterior Inference via MCMC

The Bayesian component treats model parameters as random variables and targets the *posterior distribution* via Bayes' theorem:

$$\underbrace{p(\boldsymbol{\theta} \mid \mathbf{r})}_{\text{posterior}} \propto \underbrace{p(\mathbf{r} \mid \boldsymbol{\theta})}_{\text{likelihood}} \times \underbrace{p(\boldsymbol{\theta})}_{\text{prior}}. \tag{6}$$

Rather than a single point estimate, we obtain a full distribution over plausible parameter values. This is implemented in **PyMC** using the No-U-Turn Sampler (NUTS)—a self-tuning variant of Hamiltonian Monte Carlo. Prior distributions for the VG model are specified as weakly informative:

$\mu \sim \mathcal{N}(0,\ 0.01)$ $\sigma \sim \text{HalfNormal}(0.2)$ $\nu \sim \text{Gamma}(2,\ 1)$ $\theta \sim \mathcal{N}(0,\ 0.01)$

The sampler runs 4 parallel chains of 2,000 iterations each (1,000 warm-up). Convergence is assessed via the $\hat{R}$ statistic ($\hat{R} < 1.01$ indicates convergence) and effective sample size (ESS). A key output is the **posterior predictive distribution**—the distribution of a new return r_{new} averaged over all parameter uncertainty:

$$p(r_{\text{new}} \mid \mathbf{r}) = \int_{\boldsymbol{\Theta}} f(r_{\text{new}} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{r})\, d\boldsymbol{\theta}. \tag{7}$$

This integral is approximated by Monte Carlo: for each MCMC draw $\boldsymbol{\theta}^{(s)}$, one synthetic return is generated. The resulting predictive envelope is then overlaid on the empirical histogram as a formal model check.

2.5 Model Comparison and Selection

Models are compared across multiple criteria to ensure robustness of conclusions:

Criterion	Formula / Method	Framework
Log-likelihood	$\ell(\hat{\boldsymbol{\theta}}) = \sum_i \log f(r_i)$	Frequentist
AIC	$2k - 2\ell(\hat{\boldsymbol{\theta}})$	Frequentist
BIC	$k \log n - 2\ell(\hat{\boldsymbol{\theta}})$	Frequentist
KS / Anderson-Darling	Formal goodness-of-fit test on full distribution	Frequentist
QQ plots	Visual tail assessment vs. empirical quantiles	Both
WAIC	$-2(\text{lppd} - p_{\text{WAIC}})$	Bayesian
Bayes Factors	$\mathcal{B}_{12} = p(\mathbf{r} \mid \mathcal{M}_1)/p(\mathbf{r} \mid \mathcal{M}_2)$	Bayesian

2.6 Risk Estimation: Value at Risk and Expected Shortfall

Under each fitted model, Value at Risk (VaR) and Expected Shortfall (ES) are computed at the 95% and 99% confidence levels:

$$\mathrm{VaR}_\alpha = -F^{-1}(1-\alpha), \tag{8}$$

$$\mathrm{ES}_\alpha = -\mathbb{E}[R \mid R < -\mathrm{VaR}_\alpha] = -\frac{1}{1-\alpha}\int_{-\infty}^{-\mathrm{VaR}_\alpha} x\, f(x)\, dx. \tag{9}$$

Under the Bayesian framework, both quantities are computed as posterior predictive summaries, yielding *credible intervals* rather than point estimates. This directly quantifies uncertainty in tail risk measures—a feature of direct relevance to Basel III/IV regulatory frameworks, which mandate ES at 97.5%.

Key result to demonstrate: Does the Gaussian model systematically *underestimate* VaR and ES relative to VG and NIG? How large is the gap in basis points at the 99% level, and does it widen during crisis sub-periods (2008–2009, March 2020)?

2.7 Simulation and Validation

Three simulation strategies are employed throughout the project:

Simulation Strategy 1 — Monte Carlo Path Simulation

Generate synthetic return paths under each model using fitted parameters. For the VG model, simulation proceeds via **Gamma subordination**:

1. Draw $G_i \sim \mathrm{Gamma}(\Delta t/\nu,\ \nu)$ — the stochastic clock.
2. Set $X_i = \theta G_i + \sigma\sqrt{G_i}\, Z_i$, where $Z_i \sim \mathcal{N}(0,1)$ — the return increment.
3. Compare the distribution of $\{X_i\}$ to the empirical return histogram via kernel density overlays and QQ plots.

Simulation Strategy 2 — Posterior Predictive Checks (Bayesian)

1. For each MCMC draw $\boldsymbol{\theta}^{(s)}$, simulate a synthetic dataset of n returns from $f(\cdot \mid \boldsymbol{\theta}^{(s)})$.
2. Compute summary statistics (kurtosis, 1% quantile) on each synthetic dataset.
3. Overlay the resulting distribution of statistics on the observed value. If the model is well-calibrated, the observed statistic should lie well within the predictive distribution.

Simulation Strategy 3 — Rolling VaR Backtest

1. Roll a 500-day estimation window forward over the 2000–2024 sample.
2. At each step, compute 1-day-ahead $\mathrm{VaR}_{99\%}$ under each model.
3. Count *exceedances* (days when $|r_t| > \mathrm{VaR}_{99\%}$) — a correctly specified model should produce $\approx 1\%$ violations.
4. Apply the **Kupiec (1995) likelihood ratio test** to formally assess unconditional coverage at the

5% significance level.

2.8 Data

Daily S&P 500 closing prices are sourced via the `yfinance` Python library for January 2000 – December 2024 (approximately 6,000 observations), encompassing four major stress episodes:

Sub-Period	Dates	Key Stress Event
Dot-com crash	2000–2002	Nasdaq collapse; equity bear market
GFC	2007–2009	Global financial crisis; Lehman Brothers
COVID shock	2020 Q1	Fastest 30% drawdown in S&P 500 history
Rate-hike cycle	2022–2023	Federal Reserve tightening; bond-equity correlation

Log-returns $r_t = \log(S_t/S_{t-1})$ are computed from adjusted close prices. Parameter stability is tested by fitting each model independently to each sub-period and comparing estimates.

2.9 Proposed Timeline (6–7 Weeks)

Week	Focus
1	Literature review; data collection; EDA; QQ plots; kurtosis & skewness diagnostics
2	Gaussian and Student-t MLE; benchmark VaR/ES estimates; first visualisations
3	VG model: derive likelihood, implement Bessel PDF, MLE via L-BFGS-B
4	NIG model; full AIC/BIC/KS comparison table; QQ plot panel across all models
5	Bayesian VG in PyMC: prior specification, NUTS sampling, convergence diagnostics
6	Posterior predictive checks; Bayesian VaR credible intervals; rolling backtest
7	Write-up, figures, revision

2.10 Illustrative Figure: Tail Comparison

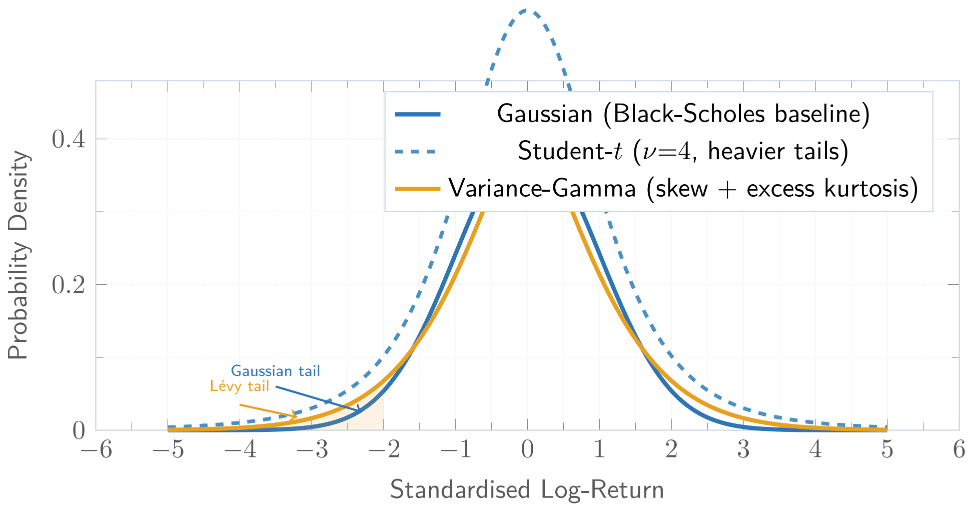

Figure 1: Schematic comparison of distributional tails. The Gaussian (solid blue) underestimates extreme negative returns. Lévy-based models (orange) assign substantially higher probability mass to tail events—the core motivation of this project.